Development of Shuttle-Tram Tracking System using IoT and Comparing of Map localization using Google APIs and Custom Map

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***Abstract*— This paper presents the development of an Internet of Things (IoT)-based Transit Tracking System. At the Faculty of Engineering, King Mongkut's Institute of Technology Ladkrabang, a public transportation system has been developed to facilitate transit around the campus using shuttle cars. Shuttle stations are placed along the routes, and cars run every 10-15 minutes. However, the problem of users not knowing the precise location and arrival time of each shuttle arose. To solve this, a transit tracking system was developed using IoT technology. IoT devices were installed on each vehicle to monitor their real-time locations. The data is collected via GPS and transmitted over the 4G network to a server, then stored in a database and displayed on a website in real-time using React. To reduce costs, a custom map was developed using the Leaflet library as an alternative to the Google Maps API. Testing within the Faculty of Engineering showed that the designed IoT system could transmit data successfully, with a performance of a custom map close to that of the Google Maps API, while also significantly reducing costs.**

Keywords—Internet of Things, Tracking System, Map Localization

I. INTRODUCTION

In large areas, public transportation systems are commonly used to reduce traffic congestion, vehicle emissions, and transportation costs. Efficient and sustainable transportation systems are essential for the smooth operation of large campuses, particularly in educational institutions where seamless mobility supports the productivity of students, faculty, and staff. In recent years, electric vehicles (EVs) have gained traction as an eco-friendly alternative to traditional fuel-based transportation, offering significant environmental and economic benefits. Within the context of campus transportation, EVs stand out for their zero-emission operation, reduced noise pollution, and lower operating costs compared to internal combustion engine vehicles. These advantages align well with the increasing focus on sustainability and energy efficiency in academic institutions, making EVs an ideal choice for shuttle services. At the School of Engineering, King Mongkut's Institute of Technology Ladkrabang (KMITL), Bangkok, Thailand, the adoption of electric shuttles offers an opportunity to further enhance the campus's environmental footprint while improving transportation reliability. A Shuttle runs every 10-15 minutes along a defined route with several stations. However, traditional methods of managing shuttle services often fail to provide real-time location updates, leading to inefficiencies such as unpredictable schedules and longer waiting times. Addressing these challenges requires the integration of innovative technologies to create a smarter, more efficient system that optimizes both the operation of EV shuttles and user experience. This prompted the development of a tracking system to monitor the real-time location of each shuttle and improve user convenience. A key aspect of this study is the comparative analysis of two mapping solutions used for data visualization: Google Maps API, a commercial mapping service, and Leaflet.js, an open-source web mapping library. The comparison examines factors such as functionality, cost-effectiveness, performance, and user-friendliness to determine the most suitable solution for campus transportation. By analyzing these mapping technologies, this research provides insights into the trade-offs between proprietary and open-source platforms for real-time tracking systems. In addition to addressing transportation inefficiencies within KMITL, the findings of this study contribute to the growing field of smart and sustainable transportation systems. The integration of IoT, GPS, cloud computing, and web-based visualization demonstrates the potential of these technologies to transform campus mobility while supporting environmental goals. Moreover, the system design presented in this research is scalable and can be adapted for use in other campuses, urban environments, or private fleet operations. The system was designed to prioritize ease of installation and use while minimizing its impact on the shuttles and reducing the number of required devices. A low-power microcontroller, commonly employed in IoT applications [1,2], was selected to collect shuttle positioning data. This choice facilitated straightforward system development and implementation. For data transmission, the 4G network was utilized [2,3], providing fast and reliable communication between the shuttle

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and the server without requiring additional infrastructure such as internet hotspots or radio towers, which are necessary in some IoT systems [1]. To enhance accessibility and usability, a web-based interface was implemented for real-time shuttle location visualization, ensuring compatibility with smartphones and other devices. Google Maps API, commonly used in similar systems [1,4], was initially employed for this purpose due to its advanced visualization capabilities. However, the cost associated with Google Maps API presented a significant consideration. To address this issue, a custom map was developed as a cost-effective alternative. This solution aims to deliver comparable performance to Google Maps API while reducing system expenses, aligning with the overall design philosophy of efficiency and scalability. The paper is organized as follows: Section 2 reviews related works. Section 3 discusses the system overview and design and implementation of the tracking system, including hardware and software architecture. Section 4 explains about a system integration, testing, and evaluation of the comparative performance of Google Maps API and Leaflet.js in terms of their functionality and cost-efficiency. Finally, Section 5 concludes with key findings, implications for future research.

II. Literature Review

The development of vehicle tracking systems utilizing GPS and IoT technologies has been extensively studied in academic research. Numerous works focus on the integration of such systems with web applications to facilitate real-time location visualization. Additionally, significant research has been conducted on the methodologies for implementing and optimizing mapping frameworks to accurately represent and identify locations within these systems. Sánchez Díaz et al. [1] proposed a transit tracking system using IoV, ITS, and machine learning to optimize routes and predict transit behavior. The system utilized a microcontroller unit (MCU) to transmit location data via LoRa and employed machine learning algorithms to enhance the accuracy of position predictions. LoRa was selected as the communication medium to minimize power consumption, but its limitations in terms of data transmission frequency and coverage necessitated the integration of machine learning. By leveraging historical location data, vehicle speed, and acceleration, the system could predict vehicle positions in near real time. Google Maps was employed for subsequent calculations and visualization. While effective for urban transit, the system lacks customization for small-scale environments like campuses. Hussain et al. [2] developed a GPS and GSM-based tracking system, highlighting its cost-effectiveness. However, GSM networks suffer from latency and inconsistent coverage, limiting scalability. To address these limitations, the proposed study employs 4G LTE for enhanced reliability and real-time data transmission. Previous works, such as Kumar et al. [3], focused on accurate GPS data collection but lacked integration with user-facing applications or custom map grids. Additionally, techniques like linear interpolation for precise mapping were not explored. The proposed study integrates interpolation to enhance location accuracy and visualization on a custom map, addressing these gaps. Mapping solutions in vehicle tracking often rely on either proprietary or open-source tools. Google Maps API, used in multiple studies, offers high functionality but is cost-prohibitive and less customizable for specific applications. Leaflet.js, explored by Alamgir et al. [4], allows for tailored mapping but requires additional development effort and lacks advanced features like traffic updates. The current study provides a comparative analysis of these two mapping technologies, tailored to a campus EV-shuttle system. While prior studies have advanced IoT-based vehicle tracking systems, limitations include reliance on GSM networks, lack of tailored mapping solutions, and limited focus on campus-specific applications. The proposed system overcomes these challenges through 4G LTE communication, a comparative evaluation of mapping technologies, and enhanced accuracy using linear interpolation. Smith et al. [5] implemented a GPS and GSM-based tracking system for urban transit but faced issues with GSM's limited coverage and latency. Zhou et al. [6] used Wi-Fi for campus shuttle tracking but found reliability challenges in outdoor environments. The proposed system overcomes these limitations with 4G LTE for faster, more reliable communication in dynamic campus scenarios. Gupta and Kumar [7] demonstrated GPS-enabled IoT tracking but did not address mapping data onto non-standard grids or the need for real-time precision. This study fills these gaps by using linear interpolation to map GPS data accurately onto a custom grid. Rahman et al. [8] highlighted the advantages of cloud computing for scalable IoT systems but noted the limitations of relying on proprietary platforms. The proposed system integrates both open-source and proprietary mapping solutions, offering a cost-effective and customizable alternative. Google Maps API [9] is widely used for its advanced features but is costly and less customizable. Chen et al. [10] demonstrated the flexibility of Leaflet.js for custom maps but noted its limited functionality for advanced features like routing. This study directly compares these two tools in terms of cost, usability, and suitability for campus-specific applications.

III. System Overview and Design

of real-time tracking and efficient management of the EV shuttle-trams operating within the campus of School of Engineering, KMITL. This section provides an overview of the system architecture and the hardware and software components used in the implementation.

A. System Architecture

The system utilizes GPS to track shuttle positions, transmitting real-time data over 4G to a server. This data is then stored in a database and made accessible through a user-friendly website, with API endpoints providing access and management to the information.

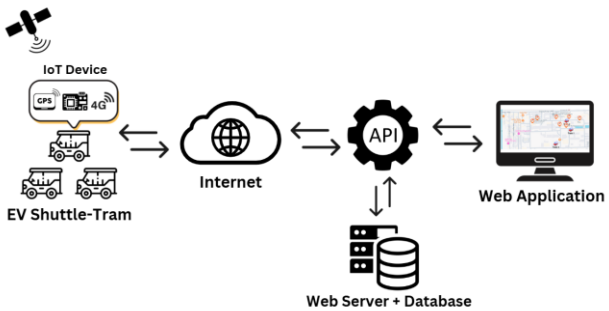


Fig. 1. System Architecture and Overview

The system uses an ESP32 microcontroller, known for its low power consumption, small size, and ease of integration with other devices and aftermarket parts. It connects to a GPS module to track the shuttle's position and

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uses a 4G module to connect to the internet to transmit the position data to the server, where data is processed and stored. The power supply is drawn from the shuttle's battery, with a power step-down to ensure the correct voltage for the system. An API was developed to handle data transmission and storage for a website that Visualized shuttle positions. This approach makes the system easier to develop and manage. Additionally, integrating an API for shuttle data reduces the complexity of the system, improves data routing efficiency, and makes it easy to develop the system. The website was built using React [11], with shuttle positions visualized on a map. Two mapping solutions were considered: the Google Maps API and a custom map developed with the Leaflet library of React [12]. To minimize costs associated with the Google Maps API.

B. IoT Hardware Device Design

The hardware installed on the shuttle includes the LILYGO T-A7670E MCU [13] with ESP32 as a core processor and integrated 4G LTE module for connecting to the internet, and an ATGM336H GPS module provides precise location coordinates (latitude and longitude) from a satellite. The system is powered by the shuttle's battery at 12V supplied, with a manual power switch to control the power on/off or can switch to a key switch if needed. This circuit is designed to be installed on each shuttle car per circuit module.

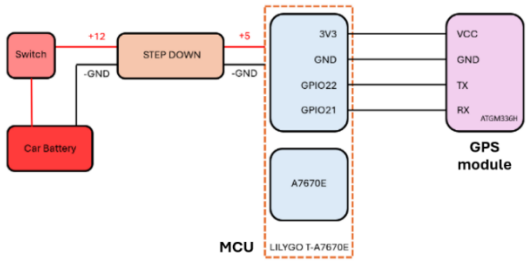


Fig. 2. Schematic diagram of IoT hardware device

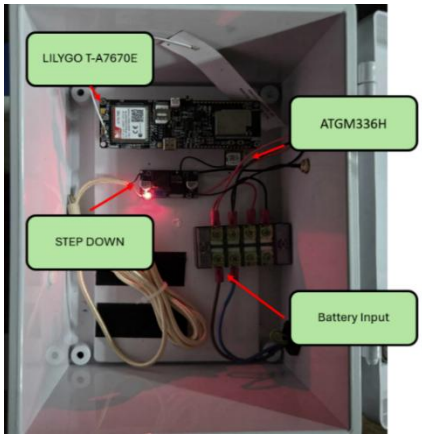


Fig. 3. IoT hardware device

The flowchart in Figure 4 illustrates the operational flow starting from the setup of each module, Starting to connect to the internet. It involves reading GPS data and transmitting it to a server via a 4G module if it can get coordinates from a satellite. Before each cycle, the system checks the 4G module's internet connectivity if the connection is lost it tries to reconnect itself and then repeats the process at this point to ensure the connection of the circuit.

C. Data Transmission and Cloud Service Design

Shuttles transmit their GPS coordinates to a server using an ExpressJS API [14]. The data is stored in a dynamic database, encompassing shuttle IDs to identify vehicles with coordinates and names to display, GPS coordinates, routes, and stations to mark the stop point of vehicles, and historical geographic location data of each vehicle for various testing purposes and future utilization. The website leverages this data to visualize shuttle locations in real time using API to manage data from the database. Comparing them against both using Google Maps API and a custom map for cost reduction, efficiency, and reliability. This functionality is enabled through API calls to retrieve data from the database and integrate it into the web application.

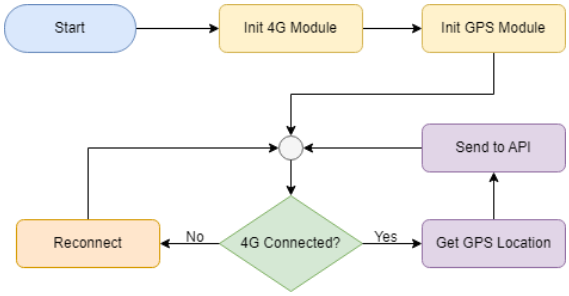


Fig. 4. Firmware of IoT device flowchart

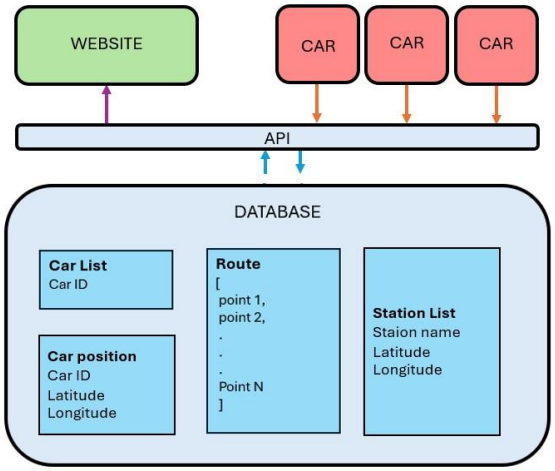


Fig. 5. System data flow

D. Web Application and Map Design

Students and staff to track the shuttle-trams in real time. It includes:Map Visualization to displays the current locations of all EV shuttles using mapping tools. Both Google Maps API and Leaflet.js are used for comparison. User Features: Includes shuttle status, estimated time of arrival (ETA), and route information. This website utilizes React to visualize the real-time shuttle positions. React supports both options, Google Maps API and the custom Leaflet-based map. An important aspect of this research is the comparative evaluation of mapping engines used to visualize shuttle locations. Two popular approaches are explored: the Google Maps API and Leaflet.js (a custom mapping solution).

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Google Maps API is a commercial mapping service that offers advanced features, including real-time traffic updates, high-resolution satellite imagery, route optimization, and seamless integration with various platforms. Its strengths lie in its reliability, extensive dataset, and user-friendly tools. However, drawbacks include dependency on Google's ecosystem, high costs for usage at scale, and limitations in customization for specific requirements. For instance, universities with a tight budget or specific geospatial needs might find these limitations restrictive. Leaflet.js, on the other hand, is an open-source JavaScript library that enables the creation of highly customizable web maps. It allows developers to tailor the mapping experience by incorporating specific layers, features, and designs that suit unique use cases. Leaflet.js is cost-effective, lightweight, and offers greater control over customization compared to Google Maps API. However, its functionality may be limited compared to Google Maps, particularly for complex routing and traffic data, and it requires more effort to integrate additional datasets or features. Leaflet-based map uses a grid overlay system to visualize positions, where actual GPS coordinates are converted into grid points for visualization in Leaflet-based map. Moreover, A responsive design to optimized for both desktop and mobile devices for easy access. The process began by isolating the geographic area corresponding to the operational service zone of the EV-shuttle tram from existing map sources. This section was then utilized to develop a customized map tailored to the specific requirements of the study. The resulting map was designed with an image resolution of 2400x2250 pixels to provide adequate detail and clarity for visualization and analytical purposes. Subsequently, the custom map was integrated with a grid system using Leaflet.js. The grid was configured such that the bottom-left corner of the map corresponded to the origin point (0,0), while the top-right corner was designated as the point (2400,2250). The grid setup is illustrated in Fig. 6.

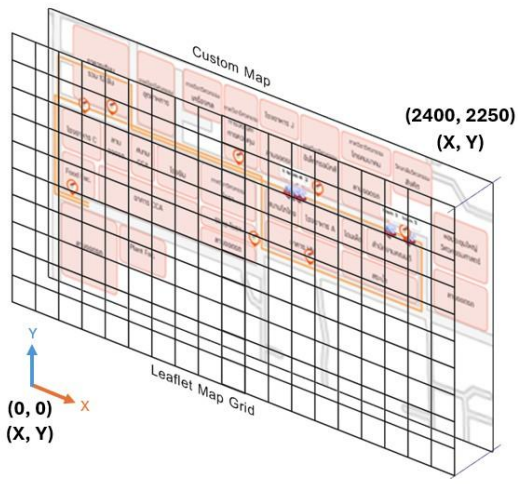


Fig. 6. Leaflet map grid and custom map

In the process of determining the shuttle tram's location for display on the web application, the IoT device determines the position and transmits the data to the server, where it is stored in the database. The web application then retrieves the location data (latitude and longitude) and vehicle information via an API. To accurately map the shuttle tram's position on the custom map, the system employs linear interpolation to transform the latitude and longitude values into corresponding coordinates on the map grid.

Then, beginning with Equation (1) for determining the scale factor by dividing the range of output values by the range of input values called a scale factor determined with (F). Where (Omax) and (Omin) represent the maximum and minimum grid point values, respectively, and (Imax) and (Imin) correspond to the maximum and minimum GPS coordinate input values. This equation will be applied identically to both Latitude and Longitude

.

$$F = \frac{O_{max} - O_{min}}{I_{max} - I_{min}} \quad (1)$$

In (2), the conversion process involves subtracting the minimum input value (I_{min}) from the coordinate to be converted. This effectively shifts the coordinate range to start from zero. Then Multiply the resulting value by the scale factor (F) calculated in Formula 1. This scales the coordinate proportionally to fit the desired grid point range. Then add the minimum output value (O_{min}) to the scaled value. This shifts the grid point value to match the desired starting point (P_{new}). This conversion process is applied identically to both Latitude and Longitude coordinates same as Formula 1.

$$P_{new} = F * (input - I_{min}) + O_{min} \quad (2)$$

Figure 7 illustrates the example of linear interpolation process for converting values by determining the ratio within a specified initial range. This ratio is then applied to scale the values to a target range. If the change in range remains constant, a fixed scale factor can be calculated and used to streamline the value conversion process.

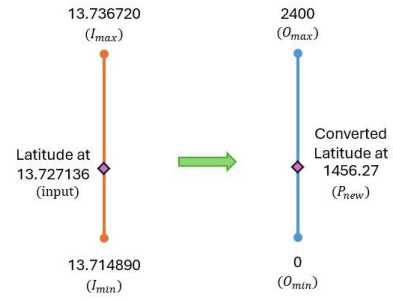


Fig. 7. Example of linear interpolation process

IV. RESULTS

The tests start from 8:30 to 16:00, September 15 to October 15, 2024, with 30 days of testing and 5 shuttles running a 10-15 minutes period testing route around the Faculty of Engineering. These tests encompassed the entire system, from the installation circuit to the shuttle car for performance and efficiency tests is record the elapsed time between the moment the MCU transmits data and the moment the server acknowledges its reception. Shuttle car position data was collected and stored in a database to assess the

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accuracy of the software components. Additionally, a comparative analysis was performed between Google Maps API and custom Leaflet maps, evaluating their performance in terms of loading times (using an online latency tester [15]) and memory usage.



Fig. 8. Tested Shuttle Car

of real-time tracking and efficient management of the EV shuttle-trams operating within the campus of School of Engineering, KMITL. This section provides an overview of the system architecture and the hardware and software components used in the implementation.

A. Hardware Testing

The hardware successfully powered and operated as expected when installed on the shuttle. The GPS module provided accurate location data with an error of distance around 2-5 meters. Data transmission from a microcontroller to a server over the 4G network test with capture timestamp, when data has to be sent from the circuit compared to timestamp when data is arriving in the database, had an average latency of 478 ms and a range of latency stayed around 320 – 950 ms, indicating stable hardware and communication.

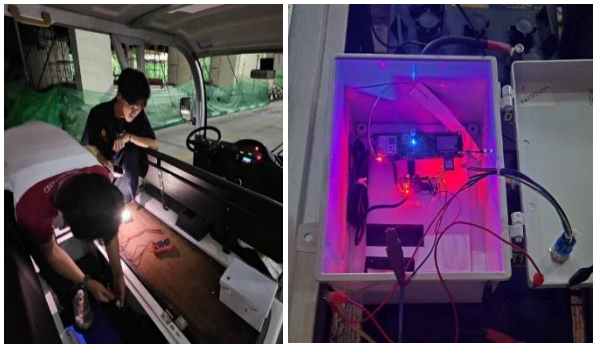


Fig. 9. Installing and Testing IoT device hardware

B. Software Testing

The API and website were rigorously tested using a custom Leaflet map in Figure 10 and Google Maps API in Figure 11 to assess their data handling and visualization capabilities. The API demonstrated seamless integration with the database, enabling efficient data transmission and management. The website successfully retrieved data from the API and presented it visually, including vehicle position pins, station markers, and route overlays on both map types. While the system performed well, future enhancements could focus on optimization, scalability, and security to ensure continued reliability and performance.



Fig. 10. Map in Website using Custom Map

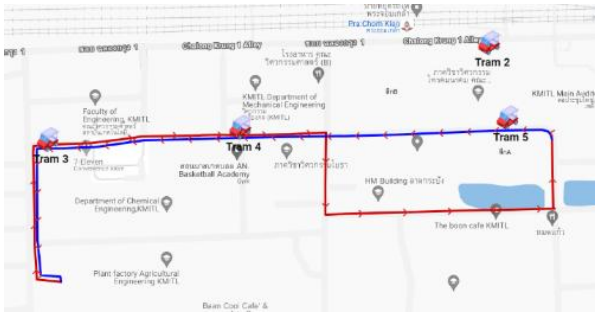


Fig. 11. Map in Website using Google Map API

To assess the accuracy of the coordinate conversion, we compared the actual coordinate data from the data store in the database to the corresponding converted coordinates used in the Custom Leaflet Map. A heatmap was generated using both sets of coordinates, and the overlapping areas of the points were analyzed to evaluate the accuracy of the conversion.

The converted coordinates were found to be slightly more contiguity than the actual coordinates shown in Figure 12, The top chart is from raw position data and the bottom is a position that is converted to use in custom Leaflet maps. This suggests that the converted coordinates offer a more accurate visualization within the test area, potential errors may arise when applying them to larger maps.

The increased accuracy of the converted coordinates can be attributed to their higher resolution. While the actual coordinates have a precision of six digits, the converted coordinates are rounded to five digits, resulting in a slightly coarser representation. However, within the specific test area, this level of precision is sufficient to provide an accurate visualization. It's important to note that this accuracy may diminish when applied to larger maps, as the cumulative effect of rounding errors could become more significant.

The custom Leaflet map required more storage space, but offered significant cost savings, reducing the API usage fees incurred with Google Maps. During the testing period, Google Maps API fees totaled 246.92 Baht, whereas the custom map incurred no additional costs. During the testing period, Google Map API usage resulted in a total cost of 246.92 baht, according to the Google Billing Report in Figure 13. The highest costs were incurred during the first 4 days of testing, likely due to the initial setup and heavy usage. Subsequently, from day 5 the costs stabilized at an average of 2-10 baht per day, indicating a more consistent usage pattern.

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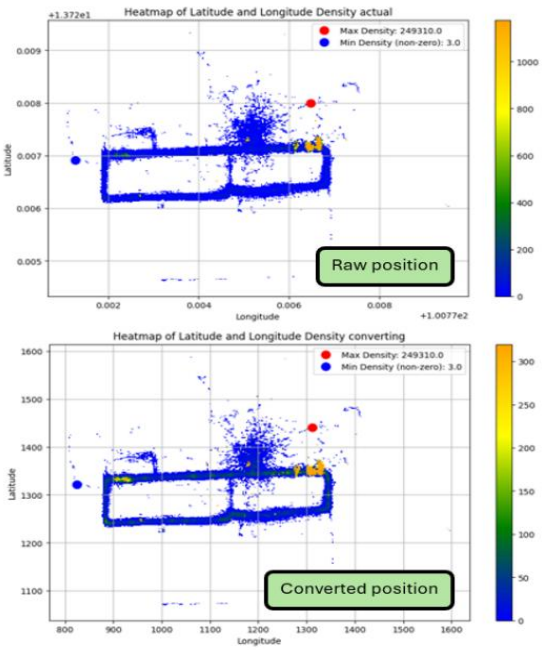


Fig. 12. Heatmap of actual and converted coordinate

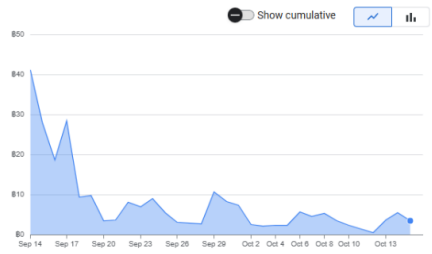


Fig. 13. Google Billing Report for Google Map API

The latency for displaying position data was tested using an online latency tester, which involved tests in 2 device configs, the result of time on both maps was similar, though Google Maps performed slightly faster on PC devices but on mobile Custom map took more time to load in first visit but better when repeat visit to website, Results are shown in Table 1.

Table 1: *Summary Latency Result*

	Latency Google Map API		Latency Custom Map	
PC	1st Visit	1.34 s	1st Visit	1.39 s
	repeat	259 ms	repeat	378 ms
Mobile	1st Visit	1.30 s	1st Visit	2.05 s
	repeat	277 ms	repeat	176 ms

To assess the memory usage between Google Maps API and Leaflet custom map, we built the React application for both options and compared their file sizes.

The results indicate that Google Maps API consumes significantly less memory than Leaflet. Google Maps API utilizes 1.24 MB, while Leaflet custom map requires 6.01 MB, representing approximately a 5x increase in memory usage.

V. Conclusion

Testing of the system's hardware and communication components demonstrated that it performed as designed. Power consumption was within specifications, individual hardware components functioned correctly, and the selected GPS module provided accurate location data. The device successfully transmitted data as intended. The choice of a 4G connection ensured reliable data transmission across various environments, as evidenced by the low latency results. A comparison of Google Maps API and a Leaflet map revealed similar latency performance, with Google Maps API exhibiting slightly lower latency in some instances. However, the custom map required significantly more storage due to the need to embed map data within the application and the inability to provide a map to be present. The primary factor influencing the decision to use a custom map was cost, as the Google Maps API incurred a charge of 246.92 baht over the 30-day testing period. Therefore, the custom map was chosen to address cost concerns and be good for long-term benefits.

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